

A Generalized Vacuum- Λ FLRW Solution and the Sign Convention of k

1. Setup

Starting line element:

$$ds^2 = A(t,r) dt^2 - a(t)^2 B(r) dr^2 - a(t)^2 C(r) r^2 d\theta^2 - a(t)^2 C(r) r^2 \sin^2(\theta) d\phi^2$$

Computing the Ricci tensor (independently verified via direct Christoffel/Riemann summation, no tensor packages) gives four non-zero components R_{00} , R_{01} , R_{11} , R_{22} .

R_{01} gives:

$$R_{01} \propto \dot{a} (\partial A / \partial r) / A = 0$$

Since $\dot{a} \neq 0$ (non-static universe), this forces $\partial A / \partial r = 0$, i.e. $A(t,r) = A(t)$, confirmed independently.

2. Reduced field equations

With $A=A(t)$, the system reduces to three equations (R_{00} , R_{11} , R_{22}) in three unknowns $A(t)$, $B(r)$, $C(r)$, satisfying $R_{\mu\nu} = \Lambda g_{\mu\nu}$ (vacuum + cosmological constant).

3. Solving R_{00} for $A(t)$

Solving R_{00} for $A(t)$ introduces an integration constant k . The correct general solution (confirmed by direct symbolic substitution) is:

$$A(t) = 3 \dot{a}^2 / (\Lambda a(t)^2 + 3|k|)$$

The absolute value is essential — without it, the formula only reproduces the correct on-shell behavior for one sign of k . The two single-sign candidates,

- $3\dot{a}^2/(\Lambda a^2 - 3k)$ ("standard" form, valid for $k \leq 0$)
- $3\dot{a}^2/(\Lambda a^2 + 3k)$ (valid for $k \geq 0$)

are each individually consistent with R_{00} reducing to $0=0$ — but that check alone is weak: R_{00} alone does not uniquely fix $A(t)$ for an arbitrary $a(t)$. It only closes once $a(t)$ additionally satisfies its own first-integral (Friedmann) constraint, and that constraint's sign is exactly what differs between the two candidates. Two formulas that are mirror images under $k \rightarrow -k$ will always both individually "check out" this way — that is not yet evidence they describe different physics.

4. Solving for B(r), C(r): the general solution

Plugging A(t) (with |k|) into R₁₁ and R₂₂ and solving algebraically:

$$B(r) = (r C'(r) + 2 C(r))^2 / [4 C(r) (r^2 |k| C(r) + 1)]$$

with C(r) left as a completely free function.

This was verified directly: substituting A(t) (with |k|) and this B(r) (with |k|) into both R¹₁ and R²₂, with C(r) kept symbolic, both equations reduce to a pure constant identity — independent of r and of the choice of C(r). This is a genuine, exactly verified solution of the field equations for arbitrary C(r), not a special case.

Full solution:

- $A(t,r) = 3\dot{a}^2 / (\Lambda a(t)^2 + 3|k|)$
- $B(t,r) = a(t)^2 (rC' + 2C)^2 / [4C(r^2|k|C + 1)]$
- $\mathcal{N}(t,r) \equiv$ angular metric factor $= a(t)^2 C(r)$
- $k \in \mathbb{R}$ free; C(r) free

This reduces to the standard FLRW metric when C(r)=1, $k \leq 0$. For $k > 0$, the C(r)=1 reduction gives $B(r) = 1/(1+kr^2)$, differing from the standard RCPC form $B(r)=1/(1-kr^2)$.

5. Resolving the $k > 0$ discrepancy: curvature invariant check

To settle whether $B(r)=1/(1+kr^2)$ (this work) and $B(r)=1/(1-kr^2)$ (standard) for the *same* k represent different physics or merely different conventions, the intrinsic Ricci scalar of the spatial 3-metric (C=1 in both cases — a coordinate-invariant quantity, unaffected by any remaining coordinate freedom) was computed directly:

- Standard metric ($B = 1/(1-kr^2)$): $R_3 = +6k$
- This work's metric ($B = 1/(1+kr^2)$): $R_3 = -6k$

Since the Ricci scalar is a curvature invariant, this is a rigorous, coordinate-independent proof: a solution labeled "k" in this work has the *same actual spatial curvature* as the standard solution labeled "-k". The two families are not different physics — they are the same solution space under the relabeling $k \rightarrow -k$.

6. Conclusion

- The generalized solution with free C(r) is a correctly-derived, independently-verified exact solution of the vacuum+ Λ field equations.
- It is **not** evidence of an error in the standard FLRW equations.
- The apparent disagreement at $k > 0$ is fully accounted for by an internal sign convention: this work's k is the negative of the standard curvature parameter.

- The genuinely useful/interesting result is the explicit demonstration — via a fully worked, independently checked exact solution — that the sign of the FLRW curvature parameter k is a convention, not a physically forced choice, together with an explicit free function $C(r)$ generalizing the radial gauge.

Appendix A: Notes on conventions used

- Metric signature: $(+,-,-,-)$, with $R_{\mu\nu} = \Lambda g_{\mu\nu}$ as the vacuum $+\Lambda$ field equation form used in the original derivation.
- All Ricci tensor computations were performed via direct Christoffel symbol and Riemann tensor summation (brute-force, no tensor-algebra package), cross-checked symbolically at each stage.

Appendix B: Reduction to RCPC via Areal Radius

This appendix shows explicitly that the free function $C(r)$ appearing in the general solution (Section 4) is pure radial gauge freedom: an exact coordinate transformation maps the general- $C(r)$ metric onto the standard RCPC form for any choice of $C(r)$, not merely $C(r)=1$.

B.1 Definition of the areal radius

Define a new radial coordinate $\rho(r) = r\sqrt{C(r)}$, chosen so that the angular part of the metric reduces to $\rho^2 d\Omega^2$ with no leftover function in front — i.e., ρ is the areal (circumference) radius.

B.2 Differential relation

$$d\rho/dr = \sqrt{C} + rC'/(2\sqrt{C}) = (2C + rC')/(2\sqrt{C}), \text{ so } dr^2 = [4C/(2C+rC')^2] d\rho^2.$$

B.3 Substitution into $B(r)dr^2$

Recall $B(r) = (rC'+2C)^2 / [4C(r^2|k|C+1)]$. Multiplying by dr^2 above, the factors $(rC'+2C)^2$ and $4C$ cancel identically, leaving $B(r)dr^2 = d\rho^2 / (r^2|k|C+1)$.

B.4 Rewriting in terms of ρ

Since $\rho^2 = r^2C$ by definition, $r^2|k|C + 1 = |k|\rho^2 + 1$, so $B(r)dr^2 = d\rho^2 / (1+|k|\rho^2)$. The angular part $C(r)r^2d\Omega^2 = \rho^2d\Omega^2$ by construction of ρ .

B.5 Result and conclusion

The full spatial line element becomes $dl^2 = d\rho^2/(1+|k|\rho^2) + \rho^2d\Omega^2$ — exactly the $B = 1/(1+kr^2)$ form from Section 5, now written in ρ for any choice of $C(r)$, not just $C(r)=1$. Every trace of $C(r)$ cancels identically; this is an exact algebraic identity, not an approximation or special case.

Combined with the Ricci-scalar invariant argument of Section 5, this establishes on two independent grounds that the free function $C(r)$ is pure radial gauge freedom, equivalent to the standard RCPC choice, and that the sign discrepancy is fully accounted for by the $k \rightarrow -k$ relabeling.

References

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